

SAFEAi

Project Team

Ahmed Hammad	21i-1543
Hiba Imran	21i-2685
Shizra Burney	21i-2660

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Supervised by

Dr. Imran Ashraf

Co-Supervised by

Dr. Salman Khalid(NESCOM)



Department of Computer Science

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

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Chapter 1

Introduction

1.1 Problem Domain

Aerial surveillance plays a crucial role in modern-day security, disaster management, and urban planning. The increasing need to monitor large, dynamic environments has led to the widespread deployment of unmanned aerial vehicles (UAVs) for tasks such as border patrol, crowd monitoring, and infrastructure management. However, the existing surveillance technologies face significant limitations, particularly in handling small object detection, tracking, and re-identification in complex environments. High altitudes, occlusion, changing lighting conditions, and weather variability further complicate effective monitoring.

Traditional surveillance systems struggle to detect, track, and re-identify small objects in real time, especially when dealing with highly congested urban areas, moving objects, and rapidly changing environments. These limitations often result in a heavy reliance on manual intervention, with large teams required to monitor and flag incidents, leading to inefficiencies and human errors.

The problem domain of this research addresses the need for more advanced, AI-driven aerial surveillance systems that can overcome the challenges posed by small object detection and tracking. The project seeks to automate the surveillance process, reduce human dependence, and ensure reliable object detection, tracking, and re-identification in various environmental conditions.

1.2 Research Problem Statement

The primary problem addressed in this research is the development of a robust AI-based system for real-time object detection, tracking, and re-identification in aerial footage, cap-

tured by UAVs operating at high altitudes. The current limitations of UAV-based surveillance include the inability to accurately detect and track small objects across multiple frames under varying conditions such as occlusions, low light, and weather changes. Moreover, the existing systems require significant computational resources and manual intervention, making them unsuitable for large-scale, continuous surveillance.

The goal of this project is to design an AI-driven system using cutting-edge deep learning techniques to enhance the accuracy and reliability of object detection and tracking. The system will incorporate advanced methodologies, such as the Slicing Aided Hyper Inference (SAHI) technique, to efficiently handle large-scale aerial images by focusing on small object detection and tracking in real time. Additionally, the system will focus on reducing the computational load while improving performance under complex real-world conditions.

1.3 Software Requirements Specification

The *SAFEAi* project involves the development of an AI-based software system for real-time object detection, tracking, and re-identification from high-altitude UAV footage. The system will leverage deep learning models and advanced computer vision techniques. The following outlines the functional and non-functional requirements necessary for its successful implementation:

1.3.1 Functional Requirements:

- **Object Detection:** The system must detect multiple objects (vehicles, people, etc.) in aerial images using YOLOv8 and Faster R-CNN algorithms, providing accurate bounding boxes around objects of interest.
- **Small Object Detection:** Implement the SAHI (Slicing Aided Hyper Inference) technique to handle large images by dividing them into patches, improving the detection accuracy of small objects.
- **Object Tracking:** Use DeepSORT for tracking multiple objects over time, maintaining object identity and tracking through frames.
- **Re-identification:** The system must re-identify objects that leave and reappear in the field of view, using CNN-based feature extraction for reliable identification across different frames and cameras.
- **Real-time Processing:** The system must provide real-time detection and tracking with minimal latency to ensure its applicability in dynamic surveillance environments.

1.3.2 Non-Functional Requirements:

- **Performance:** The software must process video streams at a minimum of 30 frames per second (FPS) to maintain real-time surveillance.
- **Scalability:** The system must scale efficiently to handle multiple UAV streams and high-resolution (up to 4K) footage.
- **Robustness:** The system must be capable of functioning effectively under varying environmental conditions such as low lighting, weather changes, and occlusions.
- **Resource Efficiency:** The software should be optimized to run on UAV-mounted hardware with limited computational resources, using lighter model versions like YOLOv4-tiny where necessary.

Chapter 2

Literature Review

This chapter critically examines existing literature relevant to the research topic. It identifies key studies, theories, and findings, providing a foundation for the proposed research and highlighting its relationship to prior work.

2.1 Related Research

2.1.1 Research Item 1: The Unmanned Aerial Vehicle Benchmark: Object Detection and Tracking (Dawei Du et al., 2018) [1]

2.1.1.1 Summary of the research item

This research presents the UAVDT dataset, a comprehensive Unmanned Aerial Vehicle (UAV) benchmark designed to advance object detection, single-object tracking (SOT), and multi-object tracking (MOT) in complex, real-world scenarios. The dataset comprises 100 video sequences, containing approximately 80,000 annotated frames selected from over 10 hours of footage captured from urban environments. The dataset is unique in its focus on unconstrained settings, which introduces challenging conditions such as varying weather, flying altitudes, camera views, and high-density traffic scenarios. A wide range of attributes—including vehicle categories, occlusions, and out-of-view conditions—are annotated to provide a detailed dataset for testing state-of-the-art computer vision algorithms. Through extensive experimentation with recent object detection and tracking algorithms, the study identifies gaps in current methods, particularly highlighting difficulties related to small object detection, real-time performance, and the challenges posed by camera motion.

2.1.1.2 Critical analysis of the research item

The UAVDT benchmark represents a significant contribution to the field of UAV-based vision systems by providing a well-annotated dataset for three fundamental tasks. The authors' efforts in annotating 80,000 frames with over 840,000 bounding boxes demonstrate the rigor and scale necessary for benchmarking. One of the strengths of the study is its focus on real-world, unconstrained scenarios, which presents more practical challenges compared to controlled datasets like those focusing on parking lots or fixed-angle cameras. The UAVDT dataset fills a gap in UAV datasets by introducing complex urban environments with multiple attributes, such as occlusions, varying altitudes, and lighting conditions, which are essential for developing robust algorithms. However, a limitation of this research is the performance of state-of-the-art algorithms on this dataset, as they struggle with small objects and high-density tracking, which suggests that existing models may need significant enhancements to handle UAV-specific challenges. Another limitation lies in the computational cost, as many accurate algorithms such as Faster R-CNN and MDP perform below real-time, making them unsuitable for UAV platforms with limited resources.

2.1.1.3 Relationship to the proposed research work

The insights gained from the UAVDT dataset are directly relevant to our proposed research, which aims to improve real-time multi-object tracking for UAV-based surveillance. The challenges identified in the UAVDT benchmark, particularly in tracking small objects and handling occlusions, align with the objectives of this study, as we aim to develop more efficient algorithms that can process complex visual data in real time. Additionally, the benchmark's inclusion of multiple object tracking and real-time constraints offers a useful testing ground for the evaluation of our proposed methodologies. The detailed analysis of the UAVDT benchmark's shortcomings also provides a clear direction for addressing gaps in current models, such as improving detection in high-density scenarios and reducing computational overhead.

2.1.2 Research Item 2: Object Detection, Recognition, and Re-identification in Video Footage (Martins E. Irhebhude, 2015) [2]

2.1.2.1 Summary of the research item

This research introduces multiple novel approaches for detecting and recognizing objects in military surveillance environments. The study focuses on recognizing military personnel, vehicles, and pedestrians, particularly in challenging settings such as night-time

surveillance or situations involving occluded or camouflaged individuals. Techniques employed in the paper include GrabCut for segmenting camouflaged personnel, Speeded Up Robust Features (SURF) for military badge recognition, and Gaussian Mixture Models (GMM) for vehicle detection. A key innovation in this work is the CENTROG descriptor, used for pedestrian and vehicle recognition in thermal images. Additionally, the study proposes methods for re-identifying personnel based on a combination of low-level and mid-level features, combining color histograms and clothing attributes to distinguish between individuals in various settings.

2.1.2.2 Critical analysis of the research item

Irhebhude's work is noteworthy for its comprehensive approach to object detection, recognition, and re-identification in military environments, tackling difficult cases such as camouflage and low-light conditions. The use of the CENTROG descriptor, a novel alternative to the widely used CENTRIST, is a key strength, particularly for improving detection in thermal imagery, which is crucial for night-time surveillance. The study's focus on multi-view object recognition is highly relevant for scenarios involving multiple UAVs or different camera angles, offering flexibility in practical applications. However, a limitation lies in the computational cost of some of the algorithms, such as the SURF-based badge recognition, which may hinder real-time performance in resource-constrained UAV systems. Furthermore, while the research effectively handles individual recognition and tracking, it does not explicitly address small object detection at high altitudes, a critical aspect for aerial surveillance that the SAFEAi project focuses on.

2.1.2.3 Relationship to the proposed research work

The methodologies explored in this research have direct relevance to the objectives of the SAFEAi project, particularly in the context of real-time object detection and re-identification in aerial surveillance. The CENTROG feature extraction method for night-time surveillance aligns with SAFEAi's requirement for reliable performance in varying lighting conditions. Additionally, the vehicle recognition techniques explored in this study can be adapted to SAFEAi's focus on small object detection, though further optimization may be required to handle the challenges of high-altitude UAV imagery. The re-identification approach based on mid-level attributes could be particularly useful for enhancing SAFEAi's tracking algorithms, enabling more precise re-identification across different frames or camera views.

2.1.3 Research Item 3: Object Tracking and Counting in a Zone using YOLOv4, DeepSORT, and TensorFlow (Kumar et al., 2021) [3]

2.1.3.1 Summary of the research item

This paper presents a system for tracking and counting objects within a specified zone using YOLOv4 for object detection and DeepSORT for tracking. The model performs multi-object detection and tracking by first identifying objects in the frame and then continuously tracking them over multiple frames. Kalman filters are applied to improve motion prediction, while DeepSORT extends the Simple Online and Realtime Tracking (SORT) algorithm by incorporating a deep association metric to reduce identity switching during tracking. The proposed system also introduces zonal counting, where objects are monitored within a predefined area, raising alarms when the count exceeds a threshold. This system has wide-ranging applications, such as traffic monitoring, managing COVID-19 restrictions, and crowd control, making it highly versatile for real-world scenarios.

2.1.3.2 Critical analysis of the research item

Kumar et al.'s work on object tracking and counting leverages state-of-the-art models to offer real-time solutions for zone-specific monitoring. The integration of YOLOv4 with DeepSORT provides a powerful combination for accurate detection and tracking, and the use of Kalman filters enhances robustness against occlusion, a major challenge in crowded environments. The performance of the system on the MOT17 dataset shows strong results, with the system achieving competitive MOTA (Multiple Object Tracking Accuracy) scores. One limitation of the research is the computational intensity required by the DeepSORT model, particularly when dealing with high-density object tracking, which may impede scalability in UAV applications. Additionally, while YOLOv4 improves detection speed and accuracy, the paper acknowledges challenges in handling adverse weather conditions or low-light environments, which are critical for UAV-based surveillance.

2.1.3.3 Relationship to the proposed research work

The methodologies in this research are highly relevant to the goals of the SAFEAi project, particularly in enhancing real-time object tracking from aerial surveillance. The zonal counting feature, along with DeepSORT's capability for persistent tracking, aligns with SAFEAi's need to monitor and track multiple small objects in complex environments, such as disaster zones or border areas. The real-time nature of the system ensures that

it can be effectively applied in UAV footage, while the robustness to occlusion complements SAFEAI's objective of overcoming environmental and situational challenges. However, the scalability concerns related to computational intensity will need to be addressed, possibly by optimizing the model for UAV resource constraints or incorporating lighter detection models like YOLOv4-tiny.

2.1.4 Research Item 4: Clustered Object Detection in Aerial Images (ICCV 2019)[5]

2.1.4.1 Summary of the research item

This paper presents the Clustered Detection (ClusDet) network, a novel approach to object detection in aerial images. The main challenge in aerial imagery is the small size and sparse distribution of objects. ClusDet addresses these challenges by using a Cluster Proposal Network (CPNet) to identify object clusters, a Scale Estimation Network (ScaleNet) to normalize object scales, and a Detection Network (DetecNet) to perform detection within these clusters. ClusDet improves detection accuracy and efficiency by focusing on clustered regions, reducing the number of regions that require detailed analysis. The system achieves state-of-the-art results on datasets like VisDrone, UAVDT, and DOTA.

2.1.4.2 Critical analysis of the research item

The Clustered Object Detection in Aerial Images paper introduces an efficient method for detecting objects in clustered regions, which reduces computational costs and improves accuracy for small objects by utilizing the Cluster Proposal Network (CPNet) and Scale Estimation Network (ScaleNet). While this approach excels in scenarios with object clustering, its reliance on clustering limits its performance in non-clustered environments. Additionally, the integration of multiple sub-networks, such as CPNet and DetecNet, increases computational complexity, which could hinder real-time processing. Despite these limitations, the method demonstrates a significant advancement in small-object detection, although it might require further optimization for real-time applications.

2.1.4.3 Relationship to the proposed research work

Our project, SAFEAI, aims to enhance high-altitude aerial surveillance by improving object detection, tracking, and re-identification. Like ClusDet, our approach also tackles the challenge of small object detection. However, SAFEAI adds a new dimension by integrating tracking and re-identification across frames. Furthermore, our proposal leverages

techniques like SAHI (Slicing Aided Hyper Inference) to efficiently handle large images, while ClusDet focuses on clustered detection. Both methods aim to improve detection performance in aerial imagery, but our approach expands this by ensuring real-time tracking and re-identification in dynamic environments.

2.1.5 Research Item 5: An Experimental Study of Vehicle Detection on Aerial Imagery Using Deep Learning-based Detection Approaches (2020) [4]

2.1.5.1 Summary of the research item

This paper compares several state-of-the-art deep learning models, including Faster R-CNN, R-FCN, and SSD, for vehicle detection in aerial imagery. The experiments are conducted on the VEDAI dataset, with Faster R-CNN (ResNet101 backbone) achieving the highest mean Average Precision (mAP) score of 39.73 percent. The paper highlights the performance trade-offs between detection accuracy and computational efficiency, noting that two-stage detectors like Faster R-CNN tend to perform better than one-stage detectors like SSD in aerial imagery tasks, particularly when handling small objects like vehicles.

2.1.5.2 Critical analysis of the research item

The paper provides a valuable comparison of deep learning models for vehicle detection in aerial imagery, emphasizing the strengths of two-stage detectors like Faster R-CNN for handling small objects. Its practical analysis helps guide model selection for similar tasks, but its focus on vehicle detection limits the broader application to other object types or more complex aerial surveillance. SSD's poor performance in small object detection highlights its limitations in low-resolution data. Despite this, the paper offers important insights that can inspire further optimization.

2.1.5.3 Relationship to the proposed research work

Our SAFEAI is directly related to the findings of this paper, as both involve aerial object detection, albeit our approach covers a broader scope. While this paper focuses on vehicle detection, SAFEAI aims to detect, track, and re-identify various objects in high-altitude surveillance, incorporating methods like SAHI to enhance small-object detection efficiency. Moreover, our approach extends beyond detection by addressing real-time tracking and re-identification, which is critical in security and surveillance operations. While the paper highlights Faster R-CNN's strength, our proposal includes YOLOv8 for

optimized real-time performance, aligning with the need for fast and accurate surveillance solutions.

2.2 Analysis Summary of Research Items

Tabular representation with discussion of the critical analysis of the research items discussed in the previous section of this chapter.

Research Item	Strengths	Weaknesses
The Unmanned Aerial Vehicle Benchmark: Object Detection and Tracking (Dawei Du et al., 2018)	- Comprehensive UAV dataset with 80,000 annotated frames. - Focus on real-world, unconstrained scenarios (varying weather, altitudes, camera views).	- Struggles with small object detection and real-time performance. - Computational cost of accurate models is high.
Object Detection, Recognition, and Re-identification in Video Footage (Martins E. Irhebhude, 2015)	- CENTROG descriptor improves detection in thermal imagery. - Multi-view object recognition offers flexibility for UAVs and different camera angles.	- Computational cost for algorithms like SURF hinders real-time performance. - Does not address small object detection at altitude.
Object Tracking and Counting in a Zone using YOLOv4, DeepSORT, and TensorFlow (Kumar et al., 2021)	- Strong performance in real-time object tracking and zonal counting. - DeepSORT reduces identity switching.	- High computational intensity for DeepSORT limits scalability. - Challenges in low-light or adverse weather conditions.
Clustered Object Detection in Aerial Images (ICCV 2019)	- CPNet and ScaleNet reduce computational costs and improve small object detection accuracy. - Strong performance in clustered object detection.	- Limited performance in non-clustered environments. - Computational complexity may hinder real-time processing.
An Experimental Study of Vehicle Detection on Aerial Imagery Using Deep Learning-based Detection Approaches (2020)	- Highlights performance trade-offs in different models for vehicle detection. - Faster R-CNN excels in small object detection.	- Focus on vehicle detection limits generalization to other object types. - SSD performs poorly with small objects and low resolution.

Table 2.1: Critical Analysis of Research Items

Chapter 3

Proposed Approach

3.0.0.1 3.1 Overview

The *SAFEAi* project aims to enhance aerial surveillance by developing an advanced AI-driven system that provides accurate object detection, tracking, and re-identification in real-time. The system addresses the key challenges posed by high-altitude surveillance, such as the difficulty in detecting small objects, variations in appearance, and environmental conditions like lighting and occlusion. The solution leverages state-of-the-art deep learning models and computer vision techniques to process UAV (Unmanned Aerial Vehicle) data efficiently and robustly.

The proposed system incorporates various AI and computer vision methodologies, including object detection algorithms, tracking models, and feature extraction techniques, to ensure reliable detection and tracking across multiple scenarios.

3.0.0.2 3.2 Proposed Solution Workflow

3.2.1 Step 1: High-Altitude Image Processing and Preprocessing The first step in our solution is capturing high-altitude aerial images and video streams from UAVs. These images are large-scale and often contain a high level of complexity. To address this, we will apply the following methods:

- **Slicing Aided Hyper Inference (SAHI):** Large-scale images are divided into smaller patches for more accurate small object detection. This method allows the system to focus on finer details in each patch, increasing detection accuracy for small objects that are difficult to identify from a distance.
- **Image Preprocessing:** The system will preprocess images by normalizing lighting conditions, removing noise, and enhancing contrast to make objects more visible for detection algorithms.



Figure 3.1: Object detection with SAHI

3.2.2 Step 2: Object Detection Once the images are preprocessed, the next step is to detect objects within the UAV-captured footage. We will utilize deep learning models designed for real-time object detection:

- **YOLOv8 (You Only Look Once version 8):** This is a fast and accurate object detection model optimized for small objects. YOLOv8 will be used to detect objects in each frame and provide bounding boxes around the detected objects.
- **Faster R-CNN (Region-based Convolutional Neural Network):** For situations where high precision is needed, such as detecting objects in crowded or occluded environments, Faster R-CNN will be employed. This model excels at handling complex object structures and multi-class detection.

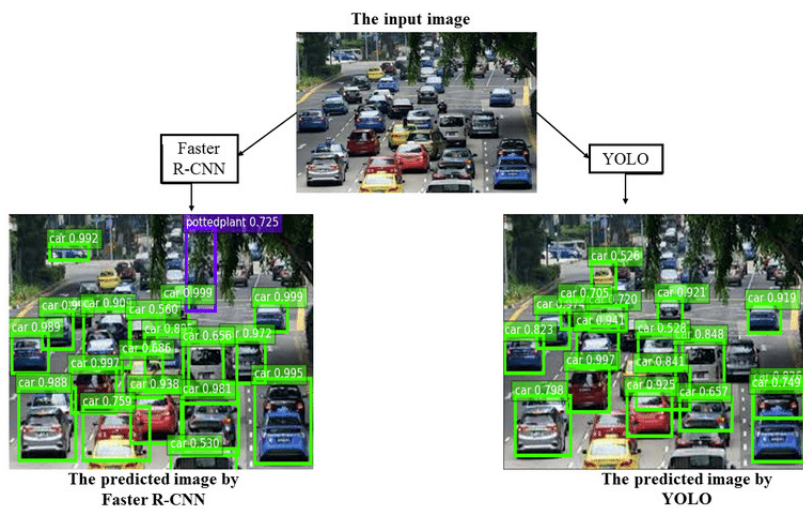


Figure 3.2: Object Detection Models: YOLOv8 and Faster R-CNN

3.2.3 Step 3: Object Tracking After detecting objects in the UAV footage, the system needs to track their movement across multiple frames to ensure continuous monitoring. We will implement the following tracking algorithms:

- **DeepSORT (Simple Online and Realtime Tracking with a Deep Association Metric):** DeepSORT will be used for real-time object tracking by assigning a unique ID to each detected object and maintaining the association of the object across frames. DeepSORT's appearance descriptor reduces identity switching, a critical feature when tracking multiple small objects in high-density areas.
- **Kalman Filters:** Kalman Filters will assist in motion prediction, helping to maintain the tracking of objects even when they are temporarily occluded or move out of the camera's field of view.

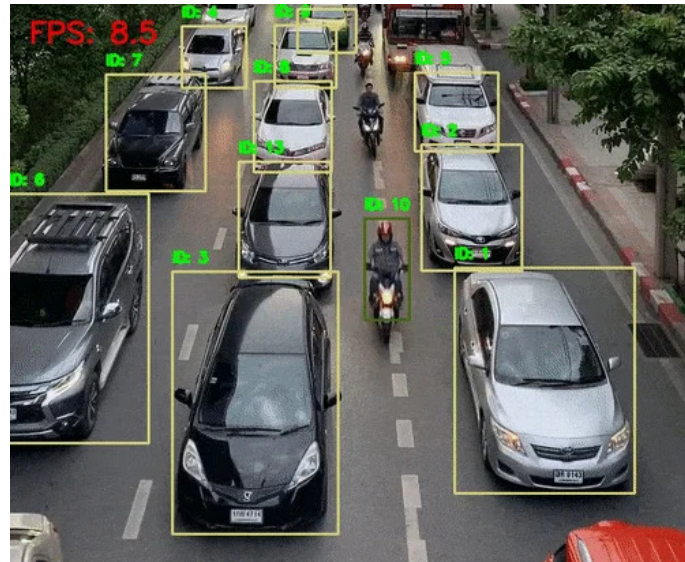


Figure 3.3: Object Tracking with Deepsort

3.2.4 Step 4: Object Re-identification Re-identification is an essential feature of the *SAFEAi* system, particularly for long-duration surveillance where objects may leave and re-enter the camera's view or appear across different UAVs. The following methods will be used for robust re-identification:

- **Convolutional Neural Networks (CNNs):** CNNs will be used for extracting distinctive features from objects (such as color, texture, and shape) to enable re-identification. Even when objects are viewed from different angles or under different lighting conditions, the CNN-based feature extractor will match them across frames.

- **Metric Learning:** A distance metric will be used to compare objects across frames and re-identify them based on their appearance features. The system will learn to recognize similarities between objects using deep metric learning techniques, minimizing identity mismatches.

3.2.5 Step 5: Real-time Processing and System Optimization Given the high demands for real-time surveillance, optimizing the system for speed and efficiency is crucial. The following strategies will be employed to achieve real-time processing:

- **Hardware Optimization:** The system will run on edge devices and UAV hardware equipped with GPUs, ensuring low-latency processing. Lighter versions of the models, such as YOLOv4-tiny, will be considered for UAVs with limited computational capacity.
- **Parallel Processing:** We will implement parallel processing on GPU cores to handle multiple image streams simultaneously, allowing for real-time surveillance across multiple UAVs.
- **Frame Skipping:** To balance the load, the system will dynamically adjust frame rates by skipping non-essential frames during low-activity periods, ensuring resources are focused on high-activity scenarios.

3.0.0.3 3.3 Challenges and Solutions

3.3.1 Small Object Detection Small object detection is a significant challenge in aerial surveillance due to high altitudes and the resolution limits of cameras. Our solution:

- Employs **SAHI** to process image patches separately, which increases detection accuracy for small objects.
- Utilizes **YOLOv8**, which is optimized for small-object detection in complex environments.

3.3.2 Real-Time Tracking Under Occlusions Objects may be occluded by buildings, trees, or other objects in the environment, making it difficult to track them consistently. To mitigate this:

- **DeepSORT** ensures identity consistency across frames, even with occlusions.
- **Kalman Filters** predict object movements during periods of occlusion, helping maintain accurate tracking.

3.3.3 Computational Resource Constraints High-altitude surveillance requires processing large images in real-time, which can be computationally expensive. To address this:

- **Efficient model architectures** like YOLOv4-tiny will be considered for resource-constrained UAV platforms.
- **SAHI** helps reduce the computational load by focusing processing efforts on critical areas of the image.

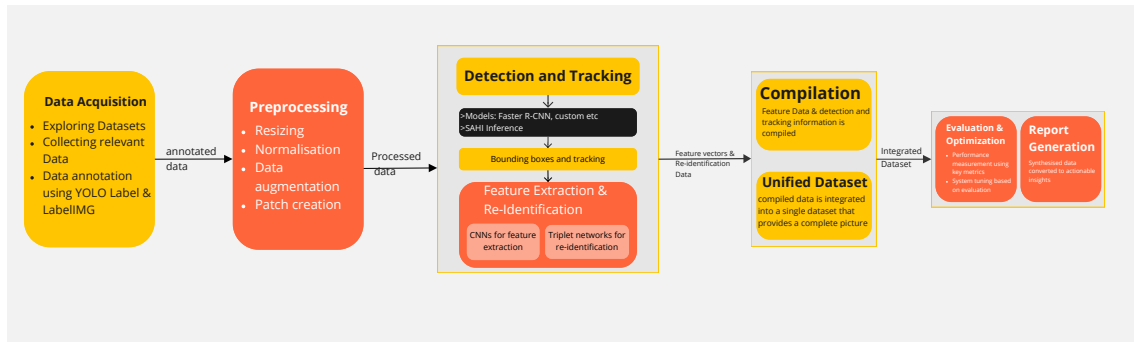


Figure 3.4: High level workflow

Chapter 4

Conclusions

The proposed solution for the *SAFEAi* project integrates advanced deep learning models with state-of-the-art object detection, tracking, and re-identification techniques to solve the challenges of high-altitude aerial surveillance. By leveraging innovative methods such as SAHI for small-object detection, DeepSORT for persistent tracking, and CNN-based feature extraction for re-identification, the system is designed to function in real-time, providing accurate, reliable surveillance even under challenging environmental conditions. This AI-driven approach ensures enhanced security and monitoring capabilities for various applications, from border security to disaster management.

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